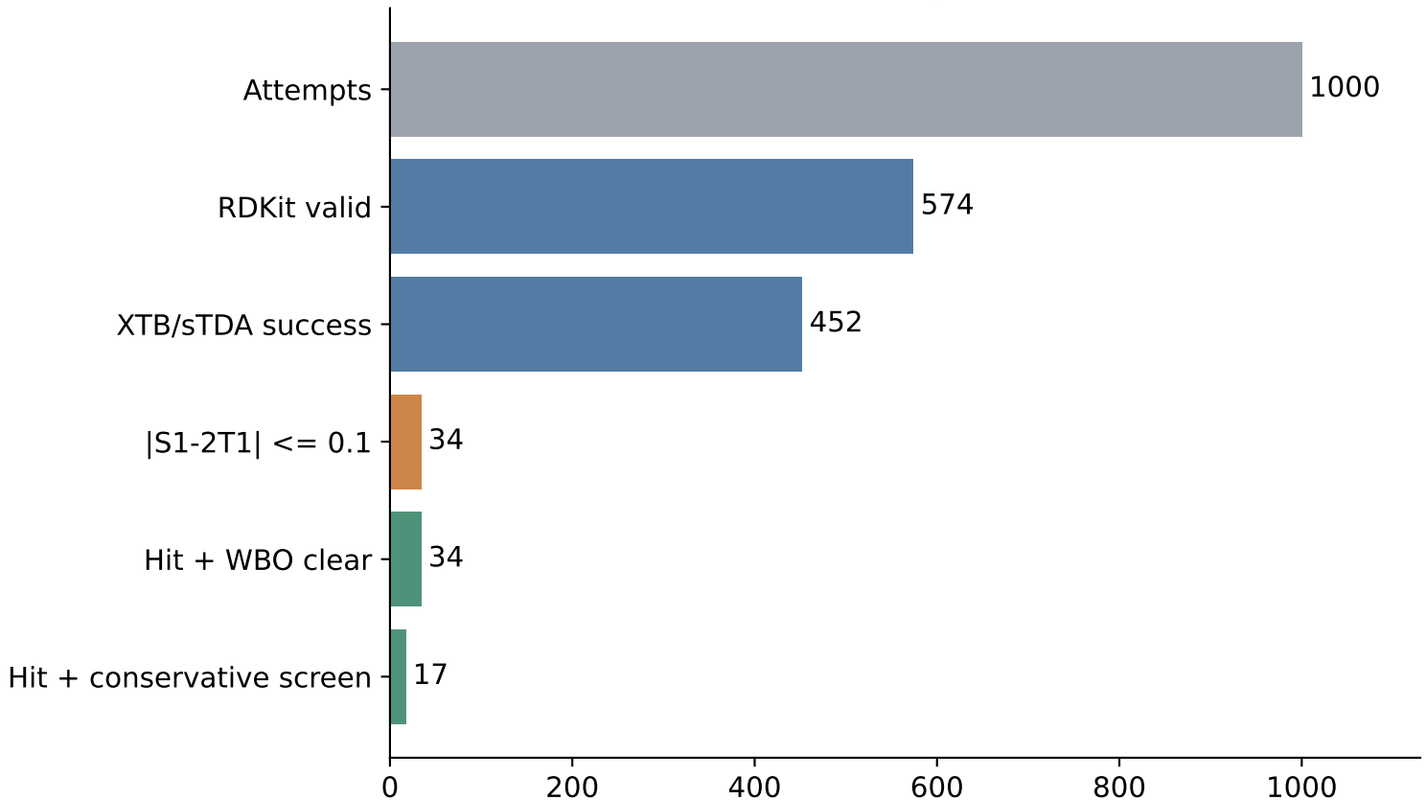
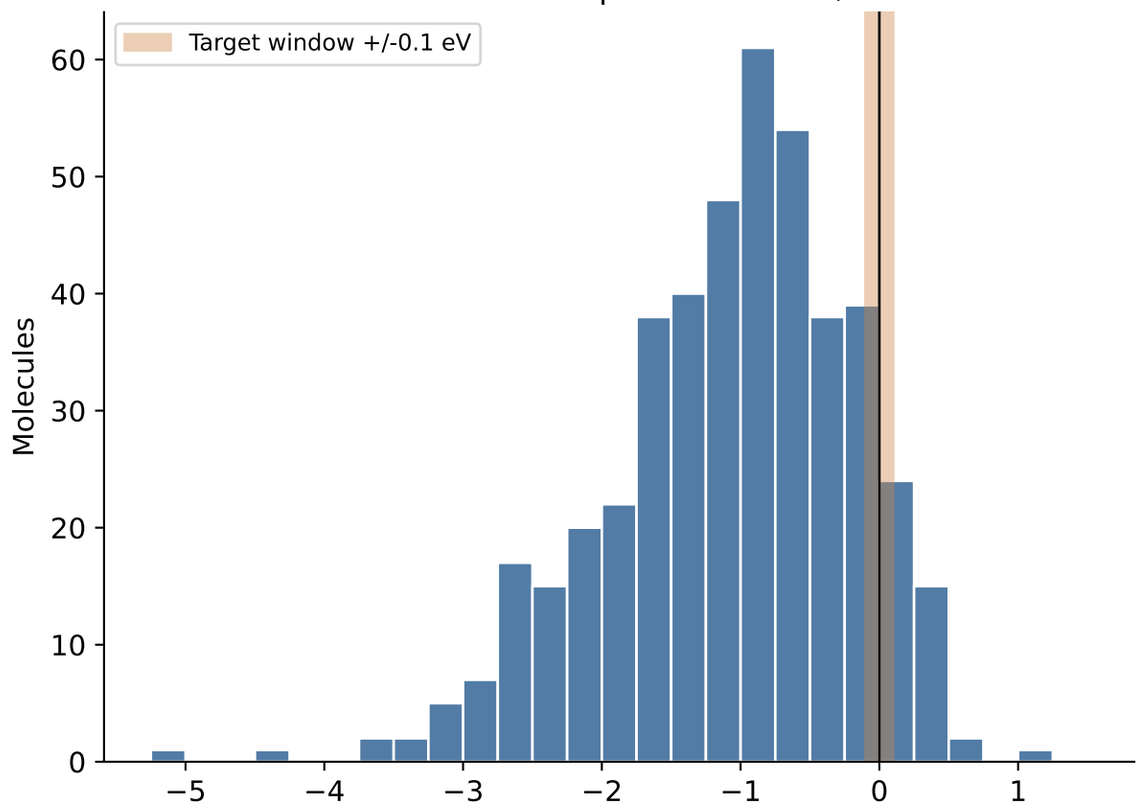
HP1KG e4690 | target S1 - 2*T1 = 0 eV | CFG 1 | 1,000 attempts

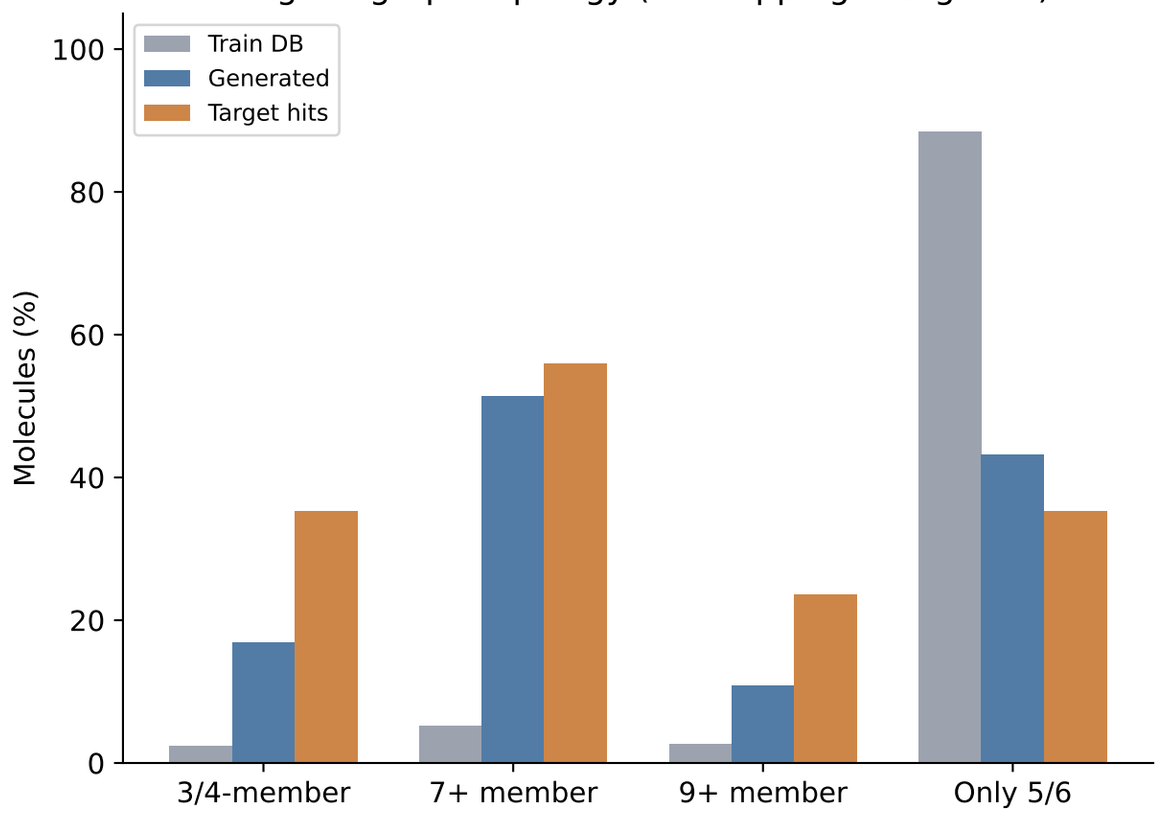
Success and target-hit funnel



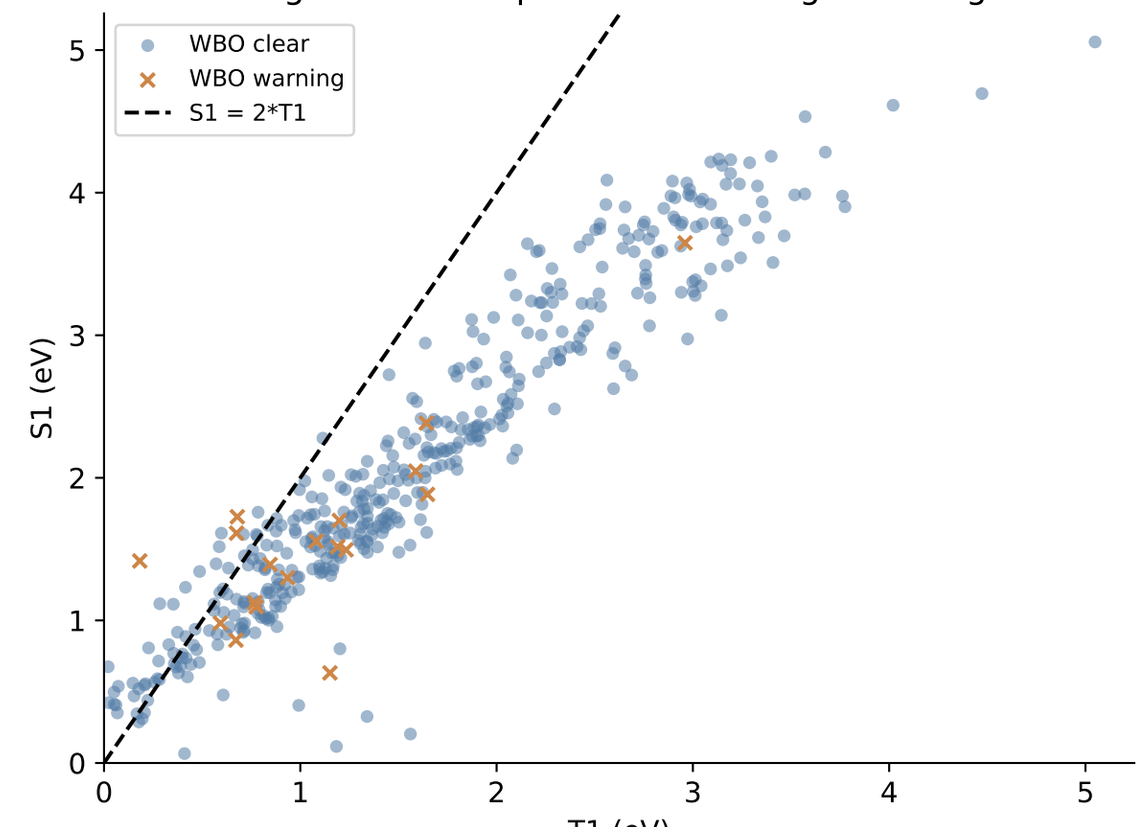
452 XTB/sTDA results | mean -1.081, std 0.880

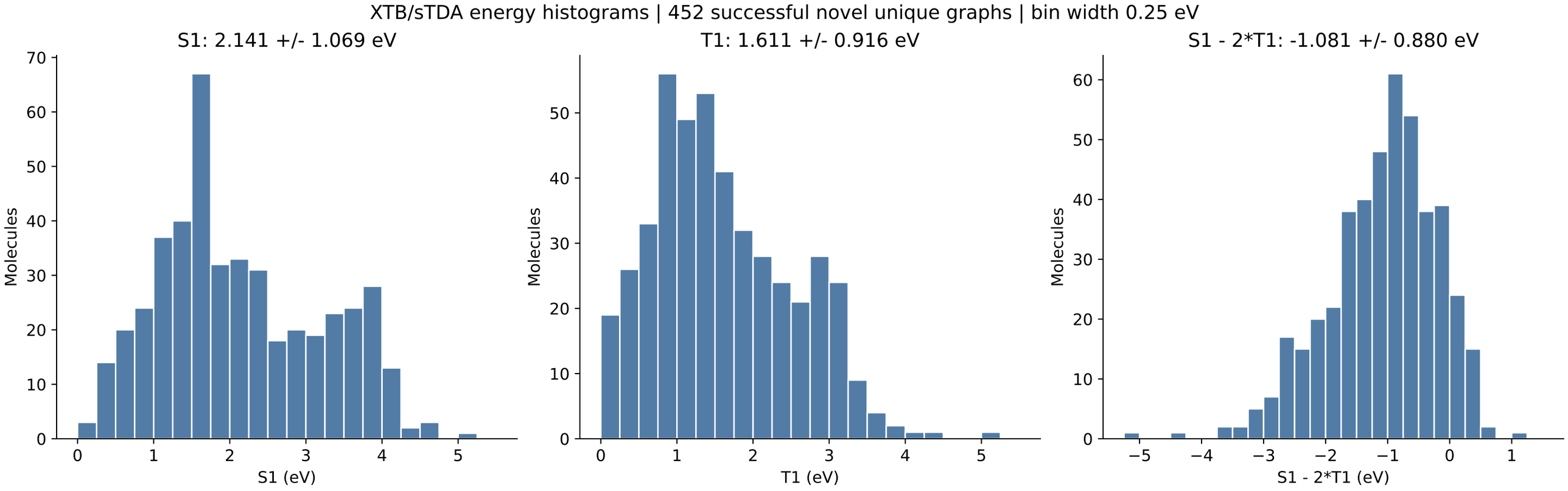


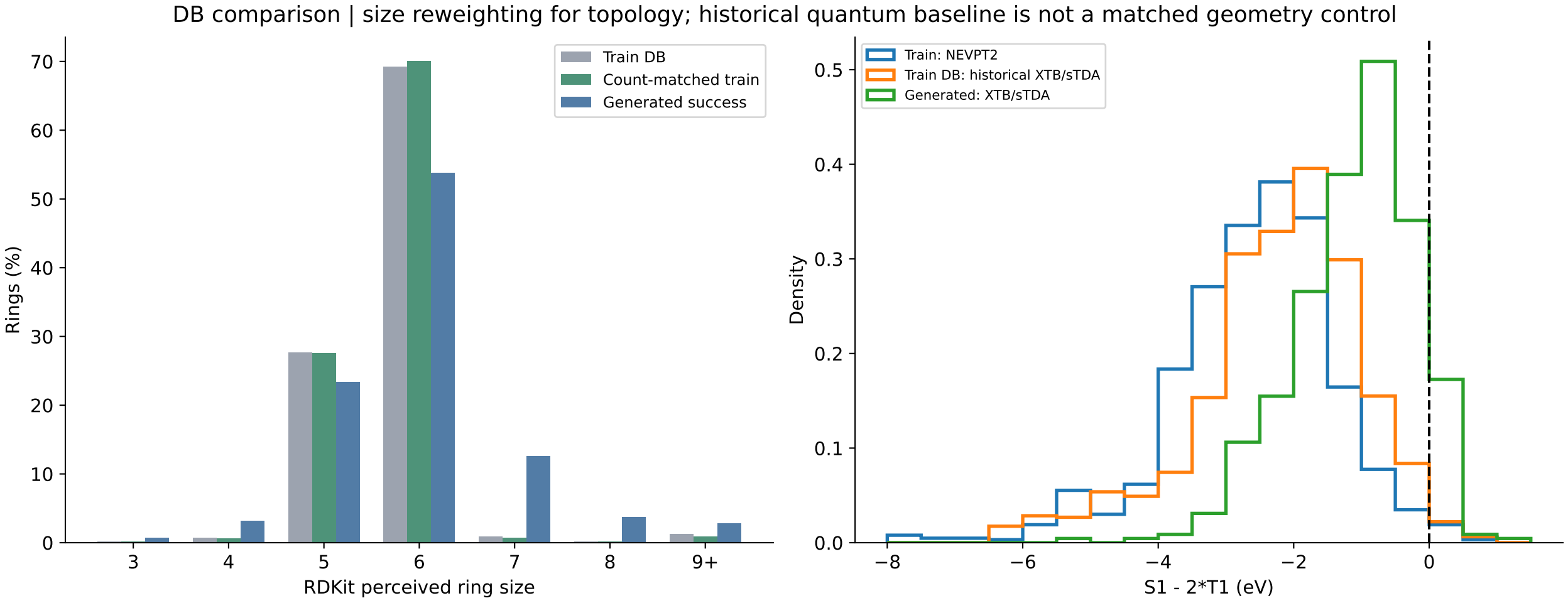
Original graph topology (overlapping categories)



Target line and optimization-change warnings




XTB/sTDA energy histograms | 452 successful novel unique graphs | bin width 0.25 eV
S1: 2.141 +/- 1.069 eV
T1: 1.611 +/- 0.916 eV
S1 - 2*T1: -1.081 +/- 0.880 eV
Molecules
S1 (eV)
T1 (eV)
S1 - 2*T1 (eV)


DB comparison | size reweighting for topology; historical quantum baseline is not a matched geometry control
Train DB
Count-matched train
Generated success
Rings (%)
RDKit perceived ring size
3
4
5
6
7
8
9+
Train: NEVPT2
Train DB: historical XTB/sTDA
Generated: XTB/sTDA
Density
S1 - 2*T1 (eV)